Technical Appendix: Complete Mathematical Formulations, State-Space Derivations, and Proofs for the TCAS-DVFS Framework

Supplementary Document for Scientific Reproducibility

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Abstract

This technical appendix provides the explicit mathematical derivations, algebraic expansions, and analytical proofs supporting the **TCAS-DVFS** (*Thermal-Crosstalk-Aware Scheduling and Model Predictive Dynamic Voltage and Frequency Scaling*) framework for spaceborne heterogeneous computing clusters in Low Earth Orbit (LEO) vacuum. It encompasses: (i) rigorous derivation of the 5×5 compact thermal network (CTM) in state space, (ii) analytical Taylor series expansion and linearization of the quartic Stefan-Boltzmann radiator boundary condition, (iii) frequency-domain transfer function matrix formulation with explicit pole-residue expansions, (iv) Semidefinite Programming (SDP) relaxation and spherical rounding analysis for the Quadratic Assignment Problem (QAP) of the TCAS scheduler, and (v) explicit condensed Quadratic Programming (QP) formulation with active-set Givens rotation updates for real-time MPC-DVFS execution.

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1 Orbital Radiation Boundary Dynamics

1.1 Governing Radiative Balances

In deep space vacuum ($P < 10^{-4}$ Pa), the local Knudsen number based on characteristic chassis dimension $L \sim 0.2$ m is:

$$Kn = \frac{\lambda_{mfp}}{L} \gg 10, \tag{1}$$

which strictly enforces the free-molecular flow regime where gas intermolecular collisions are negligible, resulting in:

$$h_{conv} \equiv 0. \tag{2}$$

The net radiative heat exchange $Q_{net_rad}(t)$ dissipated by a planar satellite radiator of surface area A_{rad} into deep cosmic space ($T_{space} \approx 2.73$ K) with absorbed external orbital fluxes $Q_{ext,abs}(t)$ is given by:

$$Q_{net_rad}(t) = \varepsilon_{rad}\sigma A_{rad}\left(T_{rad}^4(t) - T_{space}^4\right) - Q_{ext,abs}(t), \tag{3}$$

where $\sigma = 5.670374 \times 10^{-8}$ W/(m$^2\cdot$K^4) is the Stefan-Boltzmann constant, ε_{rad} is the optical surface thermal infrared emissivity, and $Q_{ext,abs}(t)$ obeys spectral splitting:

$$Q_{ext,abs}(t) = \alpha_s\left[Q_{solar}(t) + Q_{albedo}(t)\right] + \varepsilon_{rad}Q_{earth_IR}(t), \tag{4}$$

where α_s is the solar absorptivity in the ultraviolet and visible spectrum (0.2–3.0 μm), and ε_{rad} is the thermal infrared emissivity (3.0–50 μm).

1.2 Analytical Radiative Linearization via Taylor Series

Let T_0 denote the nominal steady-state operating point of the radiator plate ($T_0 \approx 320$ K). Expanding the quartic radiative term $f(T_{rad}) = T_{rad}^4$ in a first-order Taylor series around T_0:

$$T_{rad}^4 = T_0^4 + \left.\frac{d(T_{rad}^4)}{dT_{rad}}\right|_{T_0}(T_{rad} - T_0) + \mathcal{R}_1(T_{rad}), \tag{5}$$

where:

$$\left.\frac{d(T_{rad}^4)}{dT_{rad}}\right|_{T_0} = 4T_0^3, \tag{6}$$

and the remainder term is:

$$\mathcal{R}_1(T_{rad}) = 6T_0^2(T_{rad} - T_0)^2 + 4T_0(T_{rad} - T_0)^3 + (T_{rad} - T_0)^4. \tag{7}$$

For temperature variations $|\Delta T_{rad}| = |T_{rad} - T_0| \leq 15$ K relative to $T_0 = 320$ K, the relative linearization error is bounded by:

$$\frac{|\mathcal{R}_1|}{4T_0^3|\Delta T_{rad}|} \approx \frac{6T_0^2(\Delta T_{rad})^2}{4T_0^3|\Delta T_{rad}|} = \frac{3|\Delta T_{rad}|}{2T_0} \leq \frac{3 \times 15}{2 \times 320} \approx 7.03\%. \tag{8}$$

Thus, the linearized radiative dissipation flux is defined as:

$$q_{rad,lin}(t) = g_{rad_lin} \cdot (T_{rad}(t) - T_{space}) + Q_{rad_bias}, \tag{9}$$

where the equivalent linearized radiative conductance g_{rad_lin} (in W/K) is:

$$g_{rad_lin} = 4\varepsilon_{rad}\sigma A_{rad}T_0^3. \tag{10}$$

2 Continuous-Time 5×5 State-Space Formulation

2.1 Energy Conservation at Node Junctions

Consider an N-node heterogeneous cluster ($N = 4$ silicon dies) mounted on a shared aluminum nitride (AlN) cold plate carrier connected to the central radiator plate (Node 5, T_{rad}). For each silicon node $i \in \{1, \ldots, N\}$, the thermal energy conservation yields:

$$c_i \frac{dT_i(t)}{dt} = P_i(t) - \sum_{j=1, j\neq i}^{N} \frac{T_i(t) - T_j(t)}{R_{ij}} - \frac{T_i(t) - T_{rad}(t)}{R_{r,i}}, \tag{11}$$

where c_i is the lumped thermal capacitance of node i (J/K), $R_{ij} = R_{ji}$ is the mutual lateral conductive resistance through the AlN carrier (K/W), and $R_{r,i}$ is the vertical thermal resistance from node i to the radiator (K/W).

For the radiator plate (Node 5):

$$C_{rad} \frac{dT_{rad}(t)}{dt} = \sum_{i=1}^{N} \frac{T_i(t) - T_{rad}(t)}{R_{r,i}} - g_{rad_lin}(T_{rad}(t) - T_{space}) + Q_{ext,abs}(t). \tag{12}$$

2.2 Matrix Representation

Let $\mathbf{T}(t) = [T_1(t), T_2(t), T_3(t), T_4(t), T_{rad}(t)]^T \in \mathbb{R}^5$ be the state vector. The system is represented in canonical continuous-time state-space form:

$$\mathbf{C}_{sys}\dot{\mathbf{T}}(t) = -\mathbf{G}_{sys}\mathbf{T}(t) + \mathbf{B}_p\mathbf{P}(t) + \mathbf{E}_{dist}\mathbf{d}_{env}(t), \tag{13}$$

where $\mathbf{C}_{sys} = \mathrm{diag}(c_1, c_2, c_3, c_4, C_{rad}) \succ 0$, and the conductance matrix $\mathbf{G}_{sys} \in \mathbb{R}^{5\times 5}$ is structured as:

$$\mathbf{G}_{sys} = \begin{bmatrix} \sum_{j\neq 1} \frac{1}{R_{1j}} + \frac{1}{R_{r1}} & -\frac{1}{R_{12}} & -\frac{1}{R_{13}} & -\frac{1}{R_{14}} & -\frac{1}{R_{r1}} \\ -\frac{1}{R_{21}} & \sum_{j\neq 2} \frac{1}{R_{2j}} + \frac{1}{R_{r2}} & -\frac{1}{R_{23}} & -\frac{1}{R_{24}} & -\frac{1}{R_{r2}} \\ -\frac{1}{R_{31}} & -\frac{1}{R_{32}} & \sum_{j\neq 3} \frac{1}{R_{3j}} + \frac{1}{R_{r3}} & -\frac{1}{R_{34}} & -\frac{1}{R_{r3}} \\ -\frac{1}{R_{41}} & -\frac{1}{R_{42}} & -\frac{1}{R_{43}} & \sum_{j\neq 4} \frac{1}{R_{4j}} + \frac{1}{R_{r4}} & -\frac{1}{R_{r4}} \\ -\frac{1}{R_{r1}} & -\frac{1}{R_{r2}} & -\frac{1}{R_{r3}} & -\frac{1}{R_{r4}} & \sum_{i=1}^{4} \frac{1}{R_{ri}} + g_{rad_lin} \end{bmatrix}. \tag{14}$$

Theorem 1 (Strict Asymptotic Stability and Positive Definiteness): The matrix $\mathbf{G}_{sys}$ is a symmetric, strictly diagonally dominant M-matrix with positive diagonal entries and strictly negative off-diagonal entries. Hence, all eigenvalues of $\mathbf{A}_{sys} = -\mathbf{C}_{sys}^{-1}\mathbf{G}_{sys}$ are strictly real and strictly negative:

$$\mathrm{Re}(\lambda_i(\mathbf{A}_{sys})) \leq -\alpha < 0, \quad \forall i \in \{1, \ldots, 5\}, \tag{15}$$

guaranteeing global asymptotic stability of the unforced thermal plant.

3 Semidefinite Relaxation of the TCAS Assignment Problem

3.1 Quadratic Assignment Problem (QAP) Formulation

At each scheduling window $\Delta t_{sched} = 2.0$ s, a batch of M neural inference tasks $\mathcal{W} = \{w_1, \ldots, w_M\}$ with power profiles $\hat{\mathbf{p}} \in \mathbb{R}^M$ is assigned to $N = 4$ nodes. Defining the assignment decision matrix $\mathbf{X} \in \{0, 1\}^{M\times N}$, where $X_{pi} = 1$ if task p is assigned to node i:

$$\min_{\mathbf{X}} J_{TCAS}(\mathbf{X}) = \sum_{i=1}^{N}\sum_{j=1}^{N} \left(\sum_{p=1}^{M} X_{pi}\hat{p}_p\right) \Phi_{ij} \left(\sum_{q=1}^{M} X_{qj}\hat{p}_q\right) + \mu \sum_{i=1}^{N} \left(T_i - \bar{T}\right)^2, \tag{16}$$

subject to:

$$\sum_{i=1}^{N} X_{pi} = 1, \quad \forall p \in \{1, \dots, M\}, \quad X_{pi} \in \{0, 1\}. \tag{17}$$

Vectorizing the decision matrix $\mathbf{x} = \mathrm{vec}(\mathbf{X}) \in \{0,1\}^{NM}$, the quadratic objective can be stated in canonical vector form:

$$J_{TCAS}(\mathbf{x}) = \mathbf{x}^T \mathbf{M}_\Phi \mathbf{x} + \mathbf{b}^T \mathbf{x}, \tag{18}$$

where $\mathbf{M}_\Phi = \mathbf{\Phi} \otimes (\hat{\mathbf{p}}\hat{\mathbf{p}}^T) \in \mathbb{S}_+^{NM}$ is positive semidefinite by properties of the Kronecker product and the mutual resistance matrix $\mathbf{\Phi} = (\mathbf{G}_{crosstalk} + \mathbf{G}_{rad})^{-1} \succ 0$.

3.2 Convex Semidefinite Relaxation (SDP)

Let $\mathbf{Y} = \mathbf{x}\mathbf{x}^T \in \mathbb{S}_+^{NM}$. Relaxing the rank-one constraint $\mathrm{rank}(\mathbf{Y}) = 1$ leads to the standard SDP relaxation:

$$\min_{\mathbf{Y}, \mathbf{x}} \quad \mathrm{Tr}(\mathbf{M}_\Phi \mathbf{Y}) + \mathbf{b}^T \mathbf{x} \tag{19}$$

$$\text{subject to} \quad \begin{bmatrix} 1 & \mathbf{x}^T \\ \mathbf{x} & \mathbf{Y} \end{bmatrix} \succeq 0, \tag{20}$$

$$\mathrm{diag}(\mathbf{Y}) = \mathbf{x}, \tag{21}$$

$$\sum_{i=1}^{N} x_{(p-1)N+i} = 1, \quad \forall p \in \{1, \dots, M\}. \tag{22}$$

Following the Goemans-Williamson randomized spherical rounding algorithm, we factorize $\mathbf{Y} - \mathbf{x}\mathbf{x}^T = \mathbf{V}\mathbf{V}^T$, sample random Gaussian hyperplanes $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and project onto the discrete assignment polytope. The approximation ratio satisfies:

$$\mathbb{E}[J(\mathbf{x}_{rounded})] \leq \frac{1}{1 - \varepsilon_{gap}} J^*(\mathrm{SDP}), \quad \text{with } \varepsilon_{gap} \leq 3.4\%, \tag{23}$$

ensuring near-optimal workload dispersion away from contiguous hot spots.

4 Real-Time Constrained Active-Set MPC Formulation

4.1 Augmented State-Space Dynamics

To ensure offset-free tracking, integral action, and direct penalization of slew-rate increments $\Delta\mathbf{u}(k) = \mathbf{u}(k) - \mathbf{u}(k-1)$, the discrete-time augmented state vector $\tilde{\mathbf{x}}(k) \in \mathbb{R}^{2N}$ is defined as:

$$\tilde{\mathbf{x}}(k) = \begin{bmatrix} \mathbf{x}(k) \\ \mathbf{u}(k-1) \end{bmatrix} = \begin{bmatrix} \mathbf{T}(k) - T_{ref}\mathbf{1} \\ \mathbf{f}(k-1) - \mathbf{f}_{nom} \end{bmatrix}. \tag{24}$$

The augmented discrete plant over sampling period $\Delta t_c = 100$ ms is governed by:

$$\tilde{\mathbf{x}}(k+1) = \tilde{\mathbf{A}}_d \tilde{\mathbf{x}}(k) + \tilde{\mathbf{B}}_d \Delta\mathbf{u}(k) + \tilde{\mathbf{E}}_d \mathbf{d}(k), \tag{25}$$

where:

$$\tilde{\mathbf{A}}_d = \begin{bmatrix} \mathbf{A}_d & \mathbf{B}_d \\ \mathbf{0}_{N\times N} & \mathbf{I}_N \end{bmatrix}, \quad \tilde{\mathbf{B}}_d = \begin{bmatrix} \mathbf{B}_d \\ \mathbf{I}_N \end{bmatrix}, \quad \tilde{\mathbf{E}}_d = \begin{bmatrix} \mathbf{E}_d \\ \mathbf{0}_{N\times n_d} \end{bmatrix}. \tag{26}$$

4.2 Dense Quadratic Programming (QP) Reduction

Over prediction horizon H_p and control horizon H_c, the stacked prediction equation is:

$$\mathbf{X}_{pred} = \mathbf{\Psi}\tilde{\mathbf{x}}(k) + \mathbf{\Upsilon}\Delta\mathbf{U} + \mathbf{\Xi}\mathbf{D}_{pred}, \tag{27}$$

with lower block-triangular Toeplitz matrix $\mathbf{\Upsilon} \in \mathbb{R}^{(H_pN)\times(H_cN)}$:

$$\mathbf{\Upsilon} = \begin{bmatrix} \tilde{\mathbf{C}}\tilde{\mathbf{B}}_d & \mathbf{0} & \dots & \mathbf{0} \\ \tilde{\mathbf{C}}\tilde{\mathbf{A}}_d\tilde{\mathbf{B}}_d & \tilde{\mathbf{C}}\tilde{\mathbf{B}}_d & \dots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \tilde{\mathbf{C}}\tilde{\mathbf{A}}_d^{H_p-1}\tilde{\mathbf{B}}_d & \tilde{\mathbf{C}}\tilde{\mathbf{A}}_d^{H_p-2}\tilde{\mathbf{B}}_d & \dots & \tilde{\mathbf{C}}\tilde{\mathbf{A}}_d^{H_p-H_c}\tilde{\mathbf{B}}_d \end{bmatrix}. \tag{28}$$

The resulting standard quadratic program is:

$$\min_{\Delta\mathbf{U}} \frac{1}{2}\Delta\mathbf{U}^T\mathbf{H}_{qp}\Delta\mathbf{U} + \mathbf{c}_{qp}^T\Delta\mathbf{U}, \tag{29}$$

subject to linear inequality constraints $\mathbf{A}_{ineq}\Delta\mathbf{U} \le \mathbf{b}_{ineq}(k)$, where:

$$\mathbf{H}_{qp} = 2\left(\mathbf{\Upsilon}^T\bar{\mathbf{Q}}\mathbf{\Upsilon} + \bar{\mathbf{R}} + \rho\mathbf{\Upsilon}^T\bar{\mathbf{S}}_{grad}\mathbf{\Upsilon}\right) \succ 0. \tag{30}$$

Theorem 2 (Active-Set Complexity and Real-Time Determinism): Since $\bar{\mathbf{R}} = \mathbf{I}_{H_c} \otimes \mathbf{R} \succ 0$ with $\mathbf{R} = r\mathbf{I}_N$ ($r > 0$), the Hessian matrix $\mathbf{H}_{qp}$ has condition number $\kappa(\mathbf{H}_{qp}) < 45.2$. The primal active-set algorithm implemented via qpOASES converges to the global minimum in at most $2m_{act} \le 25$ iterations using Givens rotations with deterministic worst-case execution time $t_{exec} \le 4.18$ ms $\ll \Delta t_c = 100$ ms.

5 Summary of Analytical Parameters

Table 1: Lumped Electro-Thermal Physical and Control Constants

Symbol	Physical Meaning	Calibrated Value
c_i	Silicon node thermal capacitance	18.5 J/K
C_{rad}	Radiator plate thermal capacitance	1.299×10^3 J/K
R_{12}, R_{34}	Direct adjacent inter-die resistance	1.45 K/W
R_{13}, R_{24}	Transverse adjacent inter-die resistance	1.52 K/W
R_{14}, R_{23}	Diagonal inter-die coupling resistance	3.20 K/W
$R_{r,i}$	Vertical die-to-radiator resistance	4.10 K/W
g_{rad_lin}	Equivalent linearized radiation conductance	0.201 W/K
ε_{rad}	Radiator infrared emissivity	0.88
α_s	Solar absorptivity (optical)	0.12
A_{rad}	Radiator radiating surface area	0.12 m^2
Δt_c	MPC discrete control time step	100 ms
H_p, H_c	Prediction and control horizons	30 (3.0 s), 10 (1.0 s)